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Motor cables for intravascular blood pumps

Abstract

Systems and methods for providing hemodynamic support to a patient with an intravascular blood pump are disclosed. In some implementations, the blood pump includes a motor and an improved motor cable for delivering electrical power to the motor. The motor includes a stator with one or more coils. The motor cable includes one or more electrical conduits. The motor cable also includes a tail portion and a head portion. In some implementations, the head portion may have an O-shape or a C-shape. The motor cable may reduce the complexity of assembling the blood pump. For example, the motor cable may reduce the risk of shorting the one or more coils and/or the one or more electrical conduits.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority from and the benefit of U.S. Provisional Application No. 63/232,323, which was filed on Aug. 12, 2021, and is incorporated by reference herein.

TECHNICAL FIELD

(1) The present technology relates to systems and methods for providing hemodynamic support to a patient with an intravascular blood pump. In some implementations, the blood pump includes a motor and an improved motor cable for delivering electrical power to the motor.

BACKGROUND

(2) An intravascular blood pump may be inserted into a patient's blood vessel (e.g., the aorta) by means of a catheter to provide hemodynamic support. An intravascular blood pump may include an inlet area, an outlet area, a cannula, and a motor housing. During operation, blood may be drawn into one or more openings of the inlet area, channeled through the cannula, and expelled through one or more openings of the outlet area by a motor disposed within the motor housing. Failure of the motor may cause serious problems for a patient. Furthermore, even if the motor can be replaced after the blood pump has been inserted in the patient, such a replacement imposes additional risks on the patient.

BRIEF SUMMARY

(3) Systems and methods for providing hemodynamic support to a patient with an intravascular blood pump are disclosed. In some implementations, the blood pump includes a motor and an improved motor cable for delivering electrical power to the motor. The motor includes a stator with one or more coils. The motor cable includes one or more electrical conduits. The motor cable also includes a tail portion and a head portion. In some implementations, the head portion may have an O-shape or a C-shape. The motor cable may reduce the complexity of assembling the blood pump. For example, the motor cable may reduce the risk of shorting the one or more coils and/or the one or more electrical conduits.

(4) One aspect of the present disclosure relates to an intravascular blood pump comprising an inlet area having one or more openings, an outlet area having one or more openings, a passage fluidically coupling the inlet and outlet areas, a motor having a rotor and a stator, and a cable having a tail portion and a head portion. The stator of the motor includes one or more coils and is configured to generate a rotating magnetic field in response to receiving one or more electrical signals at the one or more coils. The rotating magnetic field causes the rotor to rotate. Rotation of the rotor draws blood into the one or more openings of the inlet area, channels the blood through the passage, and expels the blood through the one or more openings of the outlet area. One or more electrical conduits extend through the tail and head portions of the cable. The head portion of the cable includes one or more pads. At least one of the coils of the stator of the motor is coupled to at least one of the electrical conduits of the cable through at least one of the pads of the head portion of the cable.

(5) In some implementations, the stator has an even number of coils. In some implementations, the one or more coils and the one or more electrical conduits are coupled to the one or more pads to form a star circuit configuration. In some implementations, the one or more coils and the one or more electrical conduits are coupled to the one or more pads to form a delta circuit configuration. In some implementations, the one or more coils and the one or more electrical conduits are coupled to the one or more pads to form an open end windings circuit configuration.

(6) In some implementations, the head portion of the cable is coupled to a yoke of the motor. In some implementations, the head portion of the cable is coupled to a bearing or a bushing. In some implementations, a shaft extends through (a) an opening of the bearing or the bushing and (b) an opening of the head portion of the cable. In some implementations, the head portion of the cable is O-shaped. In some implementations, the head portion of the cable is C-shaped. In some implementations, the cable is bent near an interface between the tail portion and the head portion at an angle between 45 and 135 degrees.

(7) In some implementations, at least one of the electrical conduits includes a plurality of electrically conductive layers and a plurality of through holes, wherein each of the conductive layers is separated by at least one electrically insulating layer. In some implementations, the head portion of the cable includes an adhesive layer beneath the plurality of conductive layers. In some implementations, the tail portion includes a coating covering sections of the plurality of electrically conductive layers. In some implementations, the tail portion includes one or more pads. In such implementations, the one or more pads of the tail portion and the one or more pads of the head portion may be exposed.

(8) Another aspect of the present disclosure relates to a method for assembling an intravascular blood pump comprising a motor and a cable. The motor includes a rotor and a stator having one or more coils. The stator is configured to generate a rotating magnetic field in response to receiving one or more electrical signals at the one or more coils. The rotating magnetic field causes the rotor to rotate. The cable includes a tail portion and a head portion having one or more pads. One or more electrical conduits extend through the tail and head portions of the cable. The method comprises coupling the head portion of the cable to an internal structure of the intravascular blood pump (e.g., a yoke, a bearing, or a bushing) and coupling at least one coil of the motor to at least one of the electrical conduits through at least one of the pads.

(9) In some implementations, the head portion of the cable is O-shaped or C-shaped. In some implementations, at least one of the electrical conduits comprises a plurality of electrically conductive layers and a plurality of through holes. In such implementations, each of the conductive layers is separated by at least one electrically insulating layer. In some implementations, the head portion comprises an adhesive layer beneath the plurality of conductive layers, and coupling the head portion to the internal structure of the intravascular blood pump comprises placing the adhesive layer on the internal structure. In some implementations, coupling the at least one coil to the at least one of the electrical conduits through the at least one of the pads comprises soldering the at least one coil to the at least one of the pads.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1(a) illustrates an intravascular blood pump.
- (2) FIG. 1(b) illustrates the blood pump of FIG. 1(a) positioned within the heart of a patient.
- (3) FIG. 1(c) illustrates a ventricular support system.
- (4) FIG. 2 illustrates aspects of the interior of an intravascular blood pump.
- (5) FIGS. 3(a)-(d) illustrate coil winding patterns for individual turns in a coil.
- (6) FIGS. 3(e)-(h) illustrate coil winding patterns formed by coils having the individual turns shown in FIGS. 3(a)-(d).
- (7) FIG. 4 illustrates a cross-sectional view of a stator.
- (8) FIG. 5 illustrates a cross-sectional view of the stator of FIG. 4 as used in the blood pump of FIG. 2.
- (9) FIG. 6(a) illustrates a star or wye circuit configuration.
- (10) FIG. 6(b) illustrates a delta circuit configuration.
- (11) FIG. 6(c) illustrates an “open end windings” circuit configuration.
- (12) FIG. 6(d) illustrates a star or wye circuit configuration.
- (13) FIG. 7 illustrates a bearing.
- (14) FIGS. 8(a) and 8(b) illustrate a printed circuit board (PCB).
- (15) FIGS. 9(a) and 9(b) are illustrations of an intravascular blood pump.
- (16) FIGS. 10(a)-(c) illustrate a top, middle, and bottom electrically conductive layer, respectively, of a motor cable.
- (17) FIGS. 11(a)-(c) illustrate a top, middle, and bottom electrically conductive layer, respectively, of a motor cable.
- (18) FIG. 12 illustrates an example of a layer stack legend.
- (19) FIG. 13 is an illustration of motor cables alongside a penny.

DETAILED DESCRIPTION

(20) Implementations of the present disclosure are described in detail with reference to the drawing figures wherein like reference numerals identify similar or identical elements. It is to be understood that the disclosed implementations are merely examples of the disclosure, which may be embodied

in various forms. Well-known functions or constructions are not described in detail to avoid obscuring the present disclosure in unnecessary detail. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the present disclosure in virtually any appropriately detailed structure.

(21) An intravascular blood pump is a percutaneous, catheter-based device that can be used to provide hemodynamic support to the heart of a patient (e.g., during a high-risk percutaneous coronary intervention). As shown in FIG. 1(a), an intravascular blood pump **110** may include a pigtail **111**, an inlet area **112**, a cannula **113**, a pressure sensor **114**, an outlet area **115**, a motor housing **116**, and/or a catheter **117**. Pigtail **111** may assist with stabilizing blood pump **110** in the heart of a patient. During operation, blood may be drawn into one or more openings of inlet area **112**, channeled through cannula **113**, and expelled through one or more openings of outlet area **115** by a motor (not shown) disposed in motor housing **116**. In some implementations, the blood flow inlet and outlet areas may be reversed, such that during operation, blood may be drawn into one or more openings of outlet area **115**, channeled through cannula **113**, and expelled through one or more openings of inlet area **112**. In some implementations, pressure sensor **114** may include a flexible membrane that is integrated into cannula **113**. One side of pressure sensor **114** may be exposed to the blood pressure on the outside of cannula **113**, and the other side may be exposed to the pressure of the blood inside of cannula **113**. In some such implementations, pressure sensor **114** may generate an electrical signal proportional to the difference between the pressure outside cannula **113** and the pressure inside cannula **113**. In some implementations, a pressure difference measured by pressure sensor **114** may be used to position blood pump **110** within the heart of a patient. In some implementations, pressure sensor **114** is an optical pressure sensor. Catheter **117** may provide one or more fluidic and/or electrical connections between blood pump **110** and more or more other devices of a ventricular support system (see, e.g., FIG. 1(c)).

(22) As shown in FIG. 1(b), blood pump **110** may be positioned in a patient's heart **120**. As shown, blood pump **110** may, for example, be inserted percutaneously via the femoral artery **122** into the ascending aorta **124**, across the aortic valve **126**, and into the left ventricle **128**. In other implementations, an intravascular blood pump may, for example, be inserted percutaneously via the axillary artery **123** into the ascending aorta **124**, across the aortic valve **126**, and into the left ventricle **128**. In other implementations, an intravascular blood pump may, for example, be inserted directly into the ascending aorta **124**, across the aortic valve **126**, and into the left ventricle **128**. During operation, blood pump **110** entrains blood from the left ventricle **128** and expels blood into the ascending aorta **124**. As a result, blood pump **110** performs some of the work normally done by the patient's heart **120**. The hemodynamic effects of blood pumps may include an increase in cardiac output, improvement in coronary blood flow resulting in a decrease in LV end-diastolic pressure, pulmonary capillary wedge pressure, myocardial workload, and oxygen consumption. In some implementations, blood pump **110** may be positioned within the right side of the heart and support the right sided circulation.

(23) As shown in FIG. 1(c), blood pump **110** may be incorporated into a ventricular support system **100**. Ventricular support system **100** also includes a controller **130** (e.g., an Automated Impella Controller® from Abiomed, Inc., Danvers, Mass.), a display **140**, a purge subsystem **150**, a connector cable **160**, a plug **170**, and a repositioning unit **180**. As shown, controller **130** includes display **140**. Controller **130** monitors and controls blood pump **110**. During operation, purge subsystem **150** delivers a purge fluid to blood pump **110** through catheter **117** to prevent blood from entering the motor (not shown) within motor housing **116**. In some implementations, the purge fluid is a dextrose solution (e.g., 5% dextrose in water with 25 or 50 IU/mL of heparin). Connector cable **160** provides an electrical and/or optical connection between blood pump **110** and controller **130**. Plug **170** connects catheter **117**, purge subsystem **150**, and connector cable **160**. In some implementations, plug **170** includes a memory for storing operating parameters. Repositioning unit

180 may be used to reposition blood pump **110**.

(24) As shown, purge subsystem **150** includes a container **151**, a supply line **152**, a purge cassette **153**, a purge disc **154**, purge tubing **155**, a check valve **156**, a pressure reservoir **157**, an infusion filter **158**, and a sidearm **159**. Container **151** may, for example, be a bag or a bottle. A purge fluid is stored in container **151**. Supply line **152** provides a fluidic connection between container **151** and purge cassette **153**. Purge cassette **153** may control how the purge fluid in container **151** is delivered to blood pump **110**. For example, purge cassette **153** may include one or more valves for controlling a pressure and/or flow rate of the purge fluid. Purge disc **154** includes one or more pressure and/or flow sensors for measuring a pressure and/or flow rate of the purge fluid. As shown, controller **130** interfaces with purge cassette **153** and purge disc **154**. Purge tubing **155** provides a fluidic connection between purge disc **154** and check valve **156**. Pressure reservoir **157** provides additional filling volume and pressure during a purge fluid change. In some implementations, pressure reservoir **157** includes a flexible rubber diaphragm that provides the additional filling volume and pressure by means of an expansion chamber. Infusion filter **158** helps prevent bacterial contamination and air from entering catheter **117**. Sidearm **159** provides a fluidic connection between infusion filter **158** and plug **170**.

(25) During operation, controller **130** receives measurements from pressure sensor **114**, the motor (not shown) within motor housing **116**, and purge disc **154** and controls the motor (not shown) within motor housing **116** and purge cassette **153**. As noted above, controller **130** controls and measures a pressure and/or flow rate of a purge fluid via purge cassette **153** and purge disc **154**. During operation, after exiting purge subsystem **150** through sidearm **159**, the purge fluid is channeled through a purge lumen (not shown) within catheter **117** and plug **170**. Sensor cables (not shown) within catheter **117**, connector cable **160**, and plug **170** provide an electrical and/or optical connection between pressure sensor **114** and controller **130**. Motor cables (not shown) within catheter **117**, connector cable **160**, and plug **170** provide an electrical connection between the motor within motor housing **116** and controller **130**. During operation, controller **130** receives measurements from pressure sensor **114** through the sensor cables and controls the electrical power delivered to the motor within motor housing **116** through the motor cables. By controlling, for example, the current and/or voltage delivered to the motor within motor housing **116**, controller **130** can control the speed of the motor within motor housing **116**. In some implementations, controller **130** is connected to an external power source (e.g., a battery or an electrical outlet of a power grid). In some implementations, controller **130** comprises an internal power source (e.g., a battery). When electric power is supplied by means of a battery, a patient may be afforded a greater degree of mobility.

(26) Various modifications can be made to ventricular support system **100** and one or more of its components. For example, ventricular support system **100** can be modified to accommodate a variety of different intravascular blood pumps, such as the Impella 2.5®, Impella 5.0®, Impella 5.5®, Impella LD®, Impella RP®, and Impella CP® catheters from Abiomed, Inc., Danvers, Mass. As another example, one or more sensors may be added to blood pump **100**. For example, a second pressure sensor may be added to blood pump **100** near inlet area **112** that is configured to measure a left ventricular blood pressure. In such implementations, the second pressure sensor may operate in much the same way as pressure sensor **114**. Furthermore, in such implementations, additional sensor cables may be disposed within catheter **117**, connector cable **160**, and plug **170** to provide an electrical connection between the one or more additional sensors and controller **130**. As yet another example, one or more components of ventricular support system **100** may be separated. For example, display **140** may be incorporated into another device in communication with controller **130** (e.g., wirelessly or through one or more electrical cables). As yet another example, one or more of the sensor and/or motor cables described above can be replaced with a single electrical cable having a plurality of separate electrical conduits.

(27) FIG. 2 illustrates aspects of the interior of an intravascular blood pump **200**. In some

implementations, blood pump **200** may be structured and operated in much the same way as blood pump **110**. Furthermore, in some implementations, blood pump **200** may be incorporated into a ventricular support system, such as ventricular support system **100**. As shown, blood pump **200** comprises a motor unit **210** and a pump unit **220** arranged along a longitudinal axis **205**. Motor unit **210** comprises an electric motor including a stator **240** and a rotor **250** contained within a housing **212**. Electric power is delivered to stator **240** through electrical conduits **246** and **247** extending through a catheter **230**. In some implementations, the electric power may be provided by a controller (e.g., controller **130**) connected to blood pump **200**. In some implementations, a pressure sensor (e.g., pressure sensor **114**) may be coupled to pump unit **200**.

(28) Housing **212** comprises a proximal end **214** and a distal end **216**. Proximal end **214** of housing **212** is coupled to a distal end **234** of catheter **230**, which may comprise a flexible tube. Catheter **230** comprises a lumen **232**, which extends towards the physician (i.e., proximally) for control and operation of the blood pump **200**. In FIG. 2, stator **240** and housing **212** are depicted as separate components. However, in other implementations, stator **240** may be encapsulated within housing **212** to form a single component.

(29) Rotor **250** comprises a permanent magnet **252** that is rotationally supported about a shaft **253** within a central opening **245** of stator **240**. Magnet **252** may comprise a cylindrical permanent magnet that surrounds shaft **253** within motor unit **210**. Shaft **253** extends from motor unit **210** into pump unit **220** and facilitates rotation of an impeller **260** for the pumping of blood. In some implementations, shaft **253** may be rotationally supported by one or more contact-type bearings or bushings, such as the one illustrated in FIG. 7. In some implementations, shaft **253** may be rotationally supported by one or more non-contact-type bearings or bushings (e.g., a magnetic or hydrodynamic bearing). In some implementations, shaft **253** may be rotationally supported by a first bearing or bushing positioned at proximal end **214** of housing **212** and a second bearing or bushing positioned at distal end **216** of housing **212**. In some implementations, rotor **250** may comprise two or more permanent magnets attached to shaft **253**, or an electromagnetic magnet having its own rotor windings. Further, while FIG. 2 illustrates rotor **250** as rotatable within stator **240**, motor unit **210** may be configured such that stator **240** is held stationary about shaft **253** and rotor **250** is configured as a cylinder that rotates around stator **240**.

(30) Shaft **253** extends along the length of motor unit **210** and extends into a cylindrical housing **222** of pump unit **220**. In some implementations, shaft **253** may be hollow and comprise a lumen **254** for the passage of a guidewire, for example. The distal end of shaft **253** is coupled to an impeller **260** located within housing **222**. Interaction between stator **240** and rotor **250** of motor unit **210** generates torque in rotor **250** causing shaft **253** to rotate, which, in turn, causes impeller **260** to rotate in housing **222**. When this occurs, blood may be drawn into blood pump **200** via an axial intake opening **224** for conveyance in the axial direction, the blood issuing laterally from openings **226** and flowing axially along housing **212**. In this manner blood pump **200** generates a flow of blood within the heart of the patient. In some implementations, the blood flow may be reversed, such that during operation, blood may be drawn into openings **226**, channeled through housing **222**, and expelled through opening **224**.

(31) Stator **240** extends along the length of motor unit **210** from a proximal end **242** to a distal end **243**, and comprises coils **244** wound in a particular pattern, the details of which will be provided below. In some implementations, stator **240** may comprise six coils. In other implementations, more or fewer coils may be provided (e.g., two coils, three coils, four coils, five coils, or more than six coils). Preferably, the number of coils is even so that diametrically opposed coils may form pairs (e.g., with respect to control of a magnetic field) and be controlled simultaneously. As shown in FIG. 2, stator **240** defines opening **245** in which rotor **250** is positioned. In some implementations, stator **240** is slotless, such that the coils **244** are wound upon themselves and not onto a laminated stator core. Each of the coils **244** may have an insulating coating (not shown), and, optionally, the stator **240** may be enmolded by a synthetic epoxide resin (also not shown).

(32) As shown, motor unit **210** also comprises a yoke **213** that is contained within housing **212**. Yoke **213** carries the magnetic flux produced by the permanent magnet poles of rotor **250**. Yoke **213** may be made of a magnetic material, such as steel, or a suitable alloy, such as cobalt steel. Yoke **250** may enhance the magnetic flux, which allows for reduction of the overall diameter of the blood pump **200**. In other implementations, housing **212** may serve as yoke **213**. Since yoke **213** is the outermost component of motor unit **210**, its diameter and/or thickness limits the size of stator **240**.

(33) FIGS. 3(a)-(d) illustrate the individual winding turn structures of coil winding patterns **310-313**. However, it will be understood that a complete stator, such as stator **240** in FIG. 2, may be obtained by the axial and angular arrangement of a plurality of wire turns about a longitudinal axis of a motor unit, such as longitudinal axis **205**. FIGS. 3(e)-(h) illustrate the coil winding patterns for a complete stator for each of the coil winding types in FIGS. 3(a)-(d), respectively. The horizontal axis of each of the plots in FIGS. 3(e)-(h) represents the angular position along the circumference of the respective stator and the vertical axis represents the longitudinal length of the respective stator moving from the distal end to the proximal end of the stator.

(34) FIG. 3(a) illustrates an individual coil winding pattern **310** in which each wire **314** in the coil extends from a proximal end **321**, along the length of the coil, to a distal end **325**. At distal end **325**, wire **314** follows the external perimeter of stator for 180 degrees and returns to proximal end **321**. A complete coil winding pattern formed by coils having the turns illustrated in FIG. 3(a) is shown in FIG. 3(e).

(35) FIG. 3(b) illustrates an individual rhombic coil winding pattern **311** in which each wire **315** is arranged in a bent configuration. Unlike coil winding pattern **310** of FIG. 3(a), coil winding pattern **311** comprises one continuous wire that is wound several times over, each complete turn shifted angularly. The bent configuration of the rhombic coil winding pattern when adopted in a stator may require post-assembly of the coils of each individual phase. A complete coil winding pattern having the rhombic coil winding pattern illustrated in FIG. 3(b) is shown in FIG. 3(f).

(36) FIG. 3(c) illustrates an individual helical coil winding pattern **312** in which each wire **316** is arranged in an elliptical configuration. Coil winding pattern **312** is similar to coil winding pattern **311** of FIG. 3(b), but without the bend, which simplifies the coil winding process. The helical coil winding is a one-step winding which can be easily formed without the need for any post-assembly steps. A complete coil winding pattern having the helical coil winding pattern illustrated in FIG. 3(c) is shown in FIG. 3(g).

(37) FIG. 3(d) illustrates an individual hybrid coil winding pattern **313** that comprises a coil winding that is a mixture of coil winding pattern **310** of FIG. 3(a) and coil winding pattern **311** of FIG. 3(b). Such a hybrid coil winding allows for an optimum ratio of torque to resistance by adjusting the horizontal to vertical aspect ratio of the coil. A complete coil winding comprising the hybrid coil winding patterns illustrated in FIG. 3(d) is shown in FIG. 3(h).

(38) FIG. 4 illustrates a stator **400** comprising two coils per phase per magnet pole pair for use in a three-phase two-pole electric motor. Stator **400** may be a double-winding stator (or a four-layer coil stator). Stator **400** may employ any of coil winding patterns **310-313**. When implemented with, for example, coil winding pattern **312** of FIG. 3(c), stator **400** is a double helical winding stator similar to the complete winding illustrated in FIG. 3(g). In stator **400**, each phase A, B, and C of the three-phase electric motor comprises two coils. Thus, phase A comprises coil **410** (labelled “A1”) and coil **411** (labelled “A2”), phase B comprises coil **412** (labelled “B1”) and coil **413** (labelled “B2”), and phase C comprises coil **414** (labelled “C1”) and coil **415** (labelled “C2”). As shown, stator **400** has an inner winding comprising coils A1, B1 and C1, and an outer winding comprising coils A2, B2 and C2. In some implementations, each of coils A2, B2 and C2 in the outer winding has a greater number of turns than each of coils A1, B1 and C1 in the inner winding. As shown, each of coils **410-415** has a start point and an end point, as indicated by lead wires **420-431**. In some implementations, lead wires **420-431** are located at the proximal end of stator **400** for connectivity

with the feed lines (e.g., electrical conduits **246** and **247**). In some implementations, coils **410-415** may be formed from insulated magnet wires.

(39) In some implementations, coil A1 is formed by winding the coil from a first end **420** along the circumference of stator **400** about a 120 degree span of the coil in a first direction (e.g., anticlockwise) until the end of the span of the coil where it forms a second end **421**. After forming coil A1, the coils comprising the rest of the inner winding (i.e., coils B1 and C1) may then be formed. After coils A1, B1 and C1 are formed, coils A2, B2 and C2 may be formed. Coil A2 may be formed by winding the coil from a first end **422** along the circumference of stator **400** about a 120 degree span of the coil in a first direction (e.g., anticlockwise) until the end of the span of the coil where it forms a second end **423**. After forming coil A2, the coils comprising the rest of the outer winding (i.e., coils B2 and C2) may then be formed. This winding sequence may help reduce the overall size of stator **400**. As shown, coils **410-415** are distributed equally along the circumference of stator **400** within the inner or outer winding. However, in other implementations, coils **410-415** may be distributed unequally.

(40) FIG. 5 illustrates a cross-section **500** of blood pump **200** about line X-X' of FIG. 2 employing stator **400** of FIG. 4 in a three-phase two-pole electric motor. For clarity, the windings forming coils **410-415** are omitted from FIG. 5. As described in relation to FIG. 2, rotor **250** is in constant rotation when in use. FIG. 5 depicts the position of rotor **250** at a particular instance in time. In the illustrated position, rotor **250** produces a magnetic flux density B, and each of coils **410-415** carry a current that may be directed longitudinally (into the page or out of the page). According to Lorentz force law, the interaction between the magnetic flux density B and the longitudinal length of the current-carrying wire L in a direction perpendicular to the magnetic flux density B generates a torque T within rotor **250** for rotation thereof, governed by the equation:

$$T \propto (L \{\text{circumflex over } (z)\} \times B \{\text{circumflex over } (r)\}), \quad (1)$$

where $\{\text{circumflex over } (z)\}$ is a direction parallel to the longitudinal axis **205** of rotor **250**, $\{\text{circumflex over } (r)\}$ is a radial direction of the magnetic flux density B that is perpendicular to longitudinal axis **205** of rotor **250**, and $\times$ denotes the vector cross product. Thus, the flow of current in stator **400** causes rotation of rotor **250** about longitudinal axis **205**, which, in turn, causes a corresponding rotation of impeller **260** coupled to the distal end of shaft **253**.

(41) Additional information regarding the structure and operation of the intravascular blood pumps described above in relation to FIGS. 2-5 can be found in U.S. application Ser. No. 16/909,028, which was filed on Jun. 23, 2020, and is incorporated herein by reference. In other implementations, the interior of an intravascular blood pump may be structured differently. For example, in some implementations, coils **244** may be wound upon a laminated stator core as described in E.P. Application No. 17191940.0, which was filed on Sep. 19, 2017 and published as E.P. Publication No. 3456367 A1, and E.P. Application No. 17155078.3, which was filed on Feb. 7, 2017 and published as E.P. Publication No. 3357523 A1, both of which are incorporated herein by reference.

(42) FIGS. 6(a)-(d) illustrate circuitry configurations for a stator (e.g., stators **240** or **400**) of an intravascular blood pump (e.g., blood pumps **110** or **200**). As shown, circuit configurations **601-604** include branches **611-613**. Branch **611** includes coils **621** and **624**, which may be wound upon themselves (see, e.g., coils **410-415**). Branch **612** includes coils **622** and **625**, which may be wound upon themselves (see, e.g., coils **410-415**). Branch **613** includes coils **623** and **626**, which may be wound upon themselves (see, e.g., coils **410-415**). In some implementations, each of coils **621-626** may have a single electrical conduit (e.g., an insulated magnet wire). In other implementations, one or more of coils **621-626** may include two or more electrical conduits connected in parallel. In some implementations, one or more of circuit configurations **601-604** may include more or fewer coils (e.g., two coils, three coils, four coils, five coils, or more than six coils).

(43) As shown in FIG. 6(a), circuit configuration **601** is a star or wye circuit configuration. In circuit configuration **601**, coils **621-623** are electrically connected at node **630**. Coils **621** and **624**

are electrically connected at node **631**. Coils **622** and **625** are electrically connected at node **632**. Coils **623** and **626** are electrically connected at node **633**. Coils **624**, **625**, **626** may be electrically connected to separate electrical conduits of a motor cable (e.g., electrical conduits **246** and **247**) at nodes **634**, **635**, and **636**, respectively. During operation, three-phase electrical power may be provided by a controller (e.g., controller **130**) to a stator of an intravascular blood pump. For example, three separate signals having a 120 degree phase difference may be delivered to coils **624**, **625**, and **626** through three separate electrical conduits of a motor cable and nodes **634**, **635**, and **636**, respectively. In some implementations, these signals may be oscillating signals (e.g., alternating current (AC) signals).

(44) As shown in FIG. **6(b)**, circuit configuration **602** is a delta circuit configuration. Circuit configuration **602** includes many of the same components as circuit configuration **601**. However, circuit configuration **602** does not include node **630**. Instead, circuit configuration **602** includes nodes **641-643**. Coils **622** and **624** are electrically connected at node **641**. Coils **623** and **625** are electrically connected at node **642**. Coils **621** and **626** are electrically connected at node **643**. During operation, a controller (e.g., controller **130**) may provide three-phase electrical power to coils **621-626** through three separate electrical conduits of a motor cable (e.g., electrical conduits **246** and **247**) and nodes **641-643**. For example, a first signal may be delivered through a first conduit (not shown) of a motor cable and node **641** to coils **622** and **624**, a second signal may be delivered through a second conduit (not shown) of the motor cable and node **642** to coils **623** and **625**, and a third signal may be delivered through a third conduit (not shown) of the motor cable and node **643** to coils **621** and **626**. In some implementations, the first, second, and third signals may have a 120 degree phase difference. In some implementations, the first, second, and third signals may be oscillating signals (e.g., alternating current (AC) signals).

(45) As shown in FIG. **6(c)**, circuit configuration **603** is a configuration with open wiring ends, commonly called an “open end windings” circuit configuration. Circuit configuration **603** includes many of the same components as circuit configuration **601**. However, circuit configuration **603** does not include node **630**. Instead, circuit configuration **603** includes nodes **651-656**. During operation, a controller (e.g., controller **130**) may separately provide electrical power to each of branches **611-613** through three separate pairs of electrical conduits of a motor cable (e.g., electrical conduits **246** and **247**) and nodes **651-656**. For example, in some implementations, a controller may provide three-phase electrical power to each of branches **611-613** through three separate pairs of electrical conduits of a motor cable and nodes **651-656**. In some implementations, three separate signals having a 120 degree phase difference may be delivered to branches **611-613**. In some implementations, these signals may be oscillating signals (e.g., alternating current (AC) signals).

(46) In circuit configurations **601-603**, coils **621-626** are arranged as pairs of coils in series. For example, within branch **611**, coils **621** and **624** are arranged in series. However, in other implementations, coils **621-626** may be arranged as pairs of coils in parallel. For example, as shown in FIG. **6(d)**, circuit configuration **601** may be rearranged to form circuit configuration **604**, which is also a star or wye circuit configuration.

(47) As noted above, more or fewer coils may be included in circuit configurations **601-604**. When the number of coils is increased or decreased, the number of signals delivered to branches **611-613** may also be increased or decreased. For example, in implementations where branch **613** is removed from circuit configurations **601-604**, a controller may provide two-phase electrical power to branches **611** and **612**. In some such implementations, two separate signals having a 90 or 180 degree phase difference may be delivered to branches **611** and **612**. As another example, in implementations where branches **612** and **613** are removed from circuit configurations **601-604**, a controller may provide single-phase electrical power to branch **611**. As yet another example, in implementations where a fourth branch is added to circuit configurations **601-604**, a controller may provide four-phase electrical power to branches **611-613** and the fourth branch. In some such

implementations, four separate signals having a 90 degree phase difference may be delivered to branches **611-613** and the fourth branch.

(48) FIG. 7 illustrates a bearing **700**. As shown, bearing **700** includes a top surface **710** and a central opening **720**. In some implementations, bearing **700** may be positioned within a proximal end of a housing of a motor unit (e.g., proximal end **214** of housing **212**) of an intravascular blood pump (e.g., blood pump **200**). In some implementations, bearing **700** may be used as the yoke of a blood pump. In such implementations, bearing **700** may be made of a magnetic material, such as steel, or a suitable alloy, such as cobalt steel. In other implementations, bearing **700** may be shaped differently.

(49) FIGS. **8(a)** and **8(b)** illustrate a printed circuit board (PCB) **800**. FIG. **8(a)** illustrates a top-down view of PCB **800**. FIG. **8(b)** illustrates an exploded view of PCB **800**. As shown PCB **800** includes layers **810**, **820**, and **830** and a central opening **840**. Layer **810** includes pads **811-816**. Layer **810** may be made of an electrically conductive material (e.g., copper or silver). Layer **820** may be made of an electrically insulating material (e.g., polyimide). Layer **830** may be made of an adhesive material. Pads **811-816** of PCB **800** may be used to form the circuit configurations described above in relation to FIGS. **6(a)-(d)**. Furthermore, PCB **800** may be coupled to top surface **710** of bearing **700** through the adhesive material of layer **830**. In such implementations, central opening **840** may correspond in shape to central opening **720** of bearing **700**, and its wall thickness. However, central opening **840** may be wider than central opening **720**.

(50) As noted above, pads **811-816** may be used to form the nodes of the circuit configurations described above in relation to FIGS. **6(a)-(d)**. For example, in relation to circuit configuration **601**, pads **811-813** may be used to form nodes **631-633** and pads **814-816** may be used to form nodes **634-636**. In such implementations, node **630** may be formed without the use of PCB **800** (e.g., by twisting and soldering the appropriate wires together). As another example, in relation to circuit configuration **602**, pads **811-813** may be used to form nodes **631-633** and pads **814-816** may be used to form nodes **641-643**. In such implementations the coils (e.g., coils **244** or **410-415**) of an intravascular blood pump (e.g., blood pumps **110** or **200**) may be coupled to each other or electrical conduits of a motor cable (e.g., electrical conduits **246** and **247**). As shown, pads **814-816** are larger than pads **811-813**. In relation to circuit configurations **601** and **602**, this may be advantageous because manually soldering the coils to the electrical conduits of the motor cable may be more difficult than manually soldering the coils to each other.

(51) Various modifications can be made to PCB **800**. For example, PCB **800** may include more or fewer pads (e.g., two pads, three pads, four pads, five pads, or more than six pads) and/or layers (e.g., one layer, two layers, four layers, or more than five layers). For example, in order to implement circuit configuration **601**, an additional pad may be added to PCB **800** (e.g., for node **630**). As another example, in order to implement circuit configuration **603**, three additional pads may be added to PCB **800**. As yet another example, in order to implement circuit configuration **604**, two of pads **811-816** may be removed and one or more of the remaining pads may be enlarged. In other implementations, pads **811-816** may be shaped differently. For example, each of pads **811-816** may have the same size and/or shape. Similarly, in other implementations, PCB **800** may be shaped differently. For example, in implementations where a shaft (e.g., shaft **253**) does not extend through a bearing (e.g., bearing **700**) of an intravascular blood pump (e.g., blood pumps **110** or **200**), central opening **840** may be removed from PCB **800**. In such implementations, central opening **840** may be replaced with one or more additional electrical connections (e.g., one or more additional pads). Furthermore, in such implementations, the size and/or position of pads **811-816** may be adjusted. For example, the size of pads **811-816** may be increased to further reduce the risk of unintentionally shorting coils (e.g., coils **244** or **410-415**) and/or electrical conduits of a motor cable (e.g., electrical conduits **246** and **247**).

(52) FIGS. **9(a)** and **9(b)** illustrate aspects of the interior of an intravascular blood pump **900**. Blood pump **900** may be structured and operated in much the same way as blood pumps **110** and

200. Furthermore, blood pump **900** may be incorporated into a ventricular support system, such as ventricular support system **100**. As shown, blood pump **900** includes bearing **910**, shaft **920**, PCB **930**, coils **940**, electrical conduits **951-953**, and stator **960**. Bearing **910** may be structured in much the same way as bearing **700**. Shaft **920** may be a temporary assembly fixture that is removed during assembly. PCB **930** may be structured in much the same way as PCB **800**. Coils **940** may be structured in much the same way as **410-415**. Electrical conduits **951-953** may be included in a motor cable. Stator **960** may be structured in much the same way as stator **400**. As shown in FIG. **9(b)**, during assembly, coils **940** and electrical conduits **951-953** may be manually soldered to PCB **930**. Due to the small size of blood pump **900**, the soldering process can be difficult and there is a risk of unintentionally shorting coils **940** and/or electrical conduits **951-953**.

(53) FIGS. **10(a)-(c)** illustrate aspects of a motor cable **1000** that can be used to reduce the risk of shorting coils (e.g., coils **244 410-415**, or **940**) and/or electrical conduits (e.g., electrical conduits **246, 247**, or **951-953**) of a motor cable during the assembly of an intravascular blood pump (e.g., blood pumps **110, 200**, or **900**). As shown, cable **1000** includes a top layer **1001** of FIG. **10(a)**, a middle layer **1002** of FIG. **10(b)**, a bottom layer **1003** of FIG. **10(c)**, electrical conduits **1011-1013**, input pads **1021-1023**, output pads **1031-1033**, connection pads **1041-1044**, through holes **1051-1056** and **1061-1064**, a tail portion **1070**, a head portion **1080**, and a central opening **1081**. Layers **1001-1003** may be made of an electrically conductive material (e.g., copper or silver) and separated by one or more layers (not shown) an electrically insulating material (e.g., polyimide and/or adhesive). When assembled, top layer **1001** rests above middle layer **1002**, which in turn, rests above bottom layer **1003**. In some implementations, cable **1000** may include a coating (e.g., polyimide and/or adhesive) above top layer **1001** and/or below bottom layer **1003**. Through holes **1051-1056** and **1061-1064** may be lined and/or filled with an electrically conductive material (e.g., copper or silver) to provide electrical connections between layers **1001-1003**. In some implementations, motor cable **1000** may be a flexible ribbon cable and/or a continuous printed circuit.

(54) As shown, electrical conduit **1011** includes portions of layers **1001-1003**, input pad **1021**, output pad **1031**, and through holes **1051-1053, 1061**, and **1062**. During operation, a controller (e.g., controller **130**) may provide a first signal to one or more coils (e.g., coils **244, 410-415**, or **940**) of an intravascular blood pump (e.g., blood pumps **110, 200**, or **900**) through electrical conduit **1011**. Electrical conduit **1012** includes portions of layers **1001-1003**, input pad **1022**, output pad **1032**, and through holes **1054-1056, 1063**, and **1064**. During operation, a controller may provide a second signal to one or more coils of an intravascular blood pump through electrical conduit **1012**. Electrical conduit **1013** includes portions of layer **1001**, input pad **1023**, and output pad **1033**. During operation, a controller may provide a third signal to one or more coils of an intravascular blood pump through electrical conduit **1013**.

(55) During assembly, input pads **1021-1023** may be coupled to a connector or soldered to an electrical conduit (e.g., a portion of plug **170**) for interfacing with a controller (e.g., controller **130**). Furthermore, during assembly, output pads **1031-1033** and connection pads **1041-1044** may be used to form the circuit configurations described above in relation to FIGS. **6(a)-(d)**. For example, in relation to circuit configuration **601**, connection pads **1041-1044** may be used to form nodes **630-633** and output pads **1031-1033** may be used to form nodes **634-636**. As another example, in relation to circuit configuration **602**, connection pads **1041-1044** may be used to form nodes **631-633** and output pads **1031-1033** may be used to form nodes **641-643**. In some such implementations, one of connection pads **1041-1044** may be unused. Alternatively, in other implementations, one of connection pads **1041-1044** may be removed from cable **1000**. During assembly, cable **1000** may be bent at or near the interface between tail portion **1070** and head portion **1080** at an angle between, for example, 45 and 135 degrees.

(56) In relation to, for example, blood pump **900** of FIGS. **9(a)** and **9(b)**, cable **1000** may replace PCB **930** and electrical conduits **951-953**. For example, head portion **1080** may be viewed as replacing PCB **930** and tail portion **1070** may be viewed as replacing electrical conduits **951-953**.

In such implementations, head portion **1080** may be connected to bearing **910** through the use of an adhesive. For example, an adhesive layer (not shown) may be included in head portion **1080** beneath bottom layer **1003**. Advantageously, this substitution reduces the amount of manual soldering required to assemble blood pump **900**. For example, there is no longer a need to solder electrical conduits **951-953** to PCB **930** since tail portion **1070** and head portion **1080** are already connected to one another. Furthermore, as noted above, manually soldering coils **940** to electrical conduits **951-953** may be more difficult than manually soldering coils **940** to each other. As a result, there is an overall reduced complexity, and reduced risk of unintentionally shorting coils and/or electrical conduits. For example, eliminating the solder connection of electrical conduits **951-953** to PCB **930** reduces the risk of electrically shorting the solder joints or the electrical conduits themselves to the motor housing.

(57) Various modifications can be made to cable **1000**. For example, cable **1000** may include more or fewer pads. For example, as noted above, in order to implement circuit configuration **602**, one of connection pads **1041-1044** may be removed from cable **1000**. As another example, in order to implement circuit configuration **603**, two additional connection pads may be added to cable **1000**. As yet another example, in order to implement circuit configuration **604**, three of connection pads **1041-1044** may be removed. In other implementations, cable **1000** may include more or fewer through holes. For example, additional through holes may be added to further improve or reconfigure the electrical connection between layers **1001-1003**. Due to the overall size constraints of an intravascular blood pump, this may be more effective than increasing the size of through holes **1051-1056** and/or **1061-1064**. For example, increasing the size of through holes **1051-1056** and/or **1061-1064** may require widening tail portion **1070**. In other implementations, input pads **1021-1023**, output pads **1031-1033**, and/or connection pads **1041-1044** may be shaped differently. For example, each of input pads **1021-1023**, output pads **1031-1033**, and/or connection pads **1041-1044** may have the same size and/or shape. Similarly, in other implementations, cable **1000** may be shaped differently. For example, in implementations where a shaft (e.g., shafts **253** or **920**) does not extend through a bearing (e.g., bearing **700** or **910**) of an intravascular blood pump (e.g., blood pumps **110**, **200**, or **900**), central opening **1081** may be removed from cable **1000**. In such implementations, central opening **1181** may be replaced with one or more additional electrical connections (e.g., one or more additional pads). Furthermore, in such implementations, the size and/or position of output pads **1031-1033** and/or connection pads **1041-1044** may be adjusted. For example, the size of output pads **1031-1033** and/or connection pads **1041-1044** may be increased to further reduce the risk of unintentionally shorting coils (e.g., coils **244**, **410-415**, or **940**) and/or electrical conduits **1011-1013**. In other implementations, there may be more or fewer layers in cable **1000**. For example, cable **1000** may include one or two conductive layers, with applicable pad and through-hole configurations, or cable **1000** may include four, five or more conductive layers, with applicable pad and through-hole configurations.

(58) FIGS. **11(a)-(c)** illustrate aspects of a motor cable **1100** that can be used to reduce the risk of shorting coils (e.g., coils **244**, **410-415**, or **940**) and/or electrical conduits (e.g., electrical conduits **246**, **247**, or **951-953**) of a motor cable during the assembly of an intravascular blood pump (e.g., blood pumps **110**, **200**, or **900**). As shown, cable **1100** includes a top layer **1101** of FIG. **11(a)**, a middle layer **1102** of FIG. **11(b)**, a bottom layer **1103** of FIG. **11(c)**, electrical conduits **1111-1113**, input pads **1121-1123**, output pads **1131-1133**, connection pads **1141-1144**, through holes **1151-1156** and **1161-1163**, a tail portion **1170**, a head portion **1180**, and a central opening **1181**. Layers **1101-1103** may be made of an electrically conductive material (e.g., copper or silver) and separated by one or more layers (not shown) an electrically insulating material (e.g., polyimide and/or adhesive). When assembled, top layer **1101** rests above middle layer **1102**, which in turn, rests above bottom layer **1103**. In some implementations, cable **1100** may include a coating (e.g., polyimide and/or adhesive) above top layer **1101** and/or below bottom layer **1103**. Through holes **1151-1156** and **1161-1163** may be lined and/or filled with an electrically conductive material (e.g.,

copper or silver) to provide electrical connections between layers **1101-1103**. In some implementations, motor cable **1100** may be a flexible ribbon cable and/or a continuous printed circuit. (59) As shown, electrical conduit **1111** includes portions of layers **1101-1103**, input pad **1121**, output pad **1131**, and through holes **1151-1153** and **1161**. During operation, a controller (e.g., controller **130**) may provide a first signal to one or more coils (e.g., coils **244**, **410-415**, or **940**) of an intravascular blood pump (e.g., blood pumps **110**, **200**, or **900**) through electrical conduit **1111**. Electrical conduit **1112** includes portions of layers **1101-1103**, input pad **1122**, output pad **1132**, and through holes **1154-1156**, **1162**, and **1163**. During operation, a controller may provide a second signal to one or more coils of an intravascular blood pump through electrical conduit **1112**. Electrical conduit **1113** includes portions of layer **1101**, input pad **1123**, and output pad **1133**. During operation, a controller may provide a third signal to one or more coils of an intravascular blood pump through electrical conduit **1113**.

(60) During assembly, input pads **1121-1123** may be coupled to a connector or soldered to an electrical conduit (e.g., a portion of plug **170**) for interfacing with a controller (e.g., controller **130**). Furthermore, during assembly, output pads **1131-1133** and connection pads **1141-1144** may be used to form the circuit configurations described above in relation to FIGS. **6(a)-(d)**. For example, in relation to circuit configuration **601**, connection pads **1141-1144** may be used to form nodes **630-633** and output pads **1131-1133** may be used to form nodes **634-636**. As another example, in relation to circuit configuration **602**, connection pads **1141-1144** may be used to form nodes **631-633** and output pads **1131-1133** may be used to form nodes **641-643**. In some such implementations, one of connection pads **1141-1144** may be unused. Alternatively, in other implementations, one of connection pads **1141-1144** may be removed from cable **1100**.

(61) In comparison to head portion **1080** of cable **1000**, which has an O-shape, head portion **1180** of cable **1100** has a C-shape. In some implementations, during assembly, cable **1100** may be bent in much the same way as cable **1000**. For example, cable **1100** may be bent at or near the section of tail portion **1170** between connection pads **1141** and **1144** at an angle between, for example, 45 and 135 degrees. Alternatively, during assembly, cable **1100** may be bent at or near the interface between tail portion **1170** and head portion **1180** (e.g., the section of tail portion **1170** between connection pads **1142** and **1143**) at an angle between, for example, 45 and 135 degrees. Bending cable **1100** at this location may help reduce the overall width of an intravascular blood pump (e.g., blood pumps **110**, **200**, or **900**) and/or position cable **1100** closer to a central axis of the intravascular blood pump near the connection between a motor housing (e.g., motor housing **116**) and a catheter (e.g., catheter **117**).

(62) In relation to, for example, blood pump **900** of FIGS. **9(a)** and **9(b)**, cable **1100** may replace PCB **930** and electrical conduits **951-953**. For example, head portion **1180** may be viewed as replacing PCB **930** and tail portion **1170** may be viewed as replacing electrical conduits **951-953**. In such implementations, head portion **1180** may be connected to bearing **910** through the use of an adhesive. For example, an adhesive layer (not shown) may be included in head portion **1180** beneath bottom layer **1103**. Advantageously, this substitution reduces the amount of manual soldering required to assemble blood pump **900**. For example, there is no longer a need to solder electrical conduits **951-953** to PCB **930** since tail portion **1170** and head portion **1180** are already connected to one another. Furthermore, as noted above, manually soldering coils **940** to electrical conduits **951-953** may be more difficult than manually soldering coils **940** to each other. As a result, there is an overall reduced complexity, and reduced risk of unintentionally shorting coils and/or electrical conduits. For example, eliminating the solder connection of electrical conduits **951-953** to PCB **930** reduces the risk of electrically shorting the solder joints or the electrical conduits themselves to the motor housing.

(63) Various modifications can be made to cable **1100**. For example, cable **1100** may include more or fewer pads. For example, as noted above, in order to implement circuit configuration **602**, one of connection pads **1141-1144** may be removed from cable **1100**. As another example, in order to

implement circuit configuration **603**, two additional connection pads may be added to cable **1100**. As yet another example, in order to implement circuit configuration **604**, three of connection pads **1141-1144** may be removed. In other implementations, cable **1100** may include more or fewer through holes. For example, additional through holes may be added to further improve or reconfigure the electrical connection between layers **1101-1103**. Due to the overall size constraints of an intravascular blood pump, this may be more effective than increasing the size of through holes **1151-1156** and/or **1161-1163**. For example, increasing the size of through holes **1151-1156** and/or **1161-1163** may require widening tail portion **1170**. In other implementations, input pads **1121-1123**, output pads **1131-1133**, and/or connection pads **1141-1144** may be shaped differently. For example, each of input pads **1121-1123**, output pads **1131-1133**, and/or connection pads **1141-1144** may have the same size and/or shape. Similarly, in other implementations, cable **1100** may be shaped differently. For example, in implementations where a shaft (e.g., shafts **253** or **920**) does not extend through a bearing (e.g., bearing **700** or **910**) of an intravascular blood pump (e.g., blood pumps **110**, **200**, or **900**), central opening **1181** may be removed from cable **1100**. In such implementations, central opening **1181** may be replaced with one or more additional electrical connections (e.g., one or more additional pads). Furthermore, in such implementations, the size and/or position of output pads **1131-1133** and/or connection pads **1141-1144** may be adjusted. For example, the size of output pads **1131-1133** and/or connection pads **1141-1144** may be increased to further reduce the risk of unintentionally shorting coils (e.g., coils **244**, **410-415**, or **940**) and/or electrical conduits **1111-1113**. In other implementations, there may be more or fewer layers in cable **1100**. For example, cable **1100** may include one or two conductive layers, with applicable pad and through-hole configurations, or cable **1100** may include four, five or more conductive layers, with applicable pad and through-hole configurations.

(64) FIG. **12** illustrates an example of a layer stack legend **1200** for cables **1000** and/or **1100**. In such implementations, top layers **1001** and **1101**, middle layers **1002** and **1102**, and bottom layers **1003** and **1103** have a thickness of 2.8 millimeters. Top layers **1001** and **1101** are separated from middle layers **1002** and **1102**, respectively, by a single layer of polyimide having a thickness of 1 millimeter. Middle layers **1002** and **1102** are separated from bottom layers **1003** and **1103**, respectively, by a single layer of polyimide having a thickness of 1 millimeter and two layers of adhesive each having a thickness of 2 millimeters. A 3-millimeter coating of polyimide and adhesive is provided above top layers **1001** and **1101** and below bottom layers **1003** and **1103**. In some implementations, the coating may not be included above input pads **1021-1023** and/or **1121-1123**, output pads **1031-1033** and/or **1131-1133**, and/or connection pads **1041-1044** and/or **1141-1144**. Various modifications can be made to layer stack legend **1200**. For example, any of the specific dimensions (e.g., thickness) can be increased or decreased. As another example, layers of polyimide and/or adhesive may be added or removed. For example, one or more polyimide and/or adhesive layers may be added between top layers **1001** and **1101** and middle layers **1002** and **1102**, respectively. As another example, one or more polyimide and/or adhesive layers may be removed between middle layers **1002** and **1102** and bottom layers **1003** and **1103**, respectively.

(65) FIG. **13** illustrates cables **1310**, **1320**, and **1330** alongside a penny. Cable **1310** is a prototype of cable **1000** of FIGS. **10(a)-(c)**. Cable **1320** is a prototype of cable **1100** of FIGS. **11(a)-(c)**. Cable **1330** is a motor cable having the electrical conduits **951-953** of FIGS. **9(a)** and **9(b)**. As shown, tail portions **1312** and **1322** are covered with a coating, whereas head portions **1314** and **1324** are exposed. As shown, tail portions **1312** and **1322** have a similar width to cable **1310**. Furthermore, head portions **1314** and **1324** may have a similar size to PCB **930**.

(66) From the foregoing and with reference to the various figure drawings, those skilled in the art will appreciate that certain modifications can also be made to the present disclosure without departing from the scope of the same. While several implementations of the disclosure have been shown in the drawings, it is not intended that the disclosure be limited thereto, as it is intended that the disclosure be as broad in scope as the art will allow and that the specification be read likewise.

Therefore, the above description should not be construed as limiting, but merely as exemplifications of particular implementations. Those skilled in the art will envision other modifications within the scope and spirit of the claims appended hereto.

Claims

1. An intravascular blood pump comprising: an inlet area having one or more openings; an outlet area having one or more openings; a passage fluidically coupling the inlet and outlet areas; a motor having a rotor and a stator, wherein the stator comprises one or more coils, wherein the stator is configured to generate a rotating magnetic field in response to receiving one or more electrical signals at the one or more coils, wherein the rotating magnetic field causes the rotor to rotate, and wherein rotation of the rotor draws blood into the one or more openings of the inlet area, channels the blood through the passage, and expels the blood through the one or more openings of the outlet area; and a cable having a tail portion and a head portion, wherein one or more electrical conduits extend through the tail and head portions, wherein the head portion comprises one or more pads, and wherein at least one of the coils is coupled to at least one of the electrical conduits through at least one of the pads.
2. The intravascular blood pump of claim 1, wherein the stator has an even number of coils.
3. The intravascular blood pump of claim 1, wherein the one or more coils and the one or more electrical conduits are coupled to the one or more pads to form a star circuit configuration.
4. The intravascular blood pump of claim 1, wherein the one or more coils and the one or more electrical conduits are coupled to the one or more pads to form a delta circuit configuration.
5. The intravascular blood pump of claim 1, wherein the one or more coils and the one or more electrical conduits are coupled to the one or more pads to form an open end windings circuit configuration.
6. The intravascular blood pump of claim 1, wherein the head portion of the cable is coupled to a yoke of the motor.
7. The intravascular blood pump of claim 1, wherein the head portion of the cable is coupled to a bearing or a bushing.
8. The intravascular blood pump of claim 7, wherein a shaft extends through (a) an opening of the bearing or the bushing and (b) an opening of the head portion of the cable.
9. The intravascular blood pump of claim 1, wherein at least one of the electrical conduits comprises a plurality of electrically conductive layers and a plurality of through holes, and wherein each of the conductive layers is separated by at least one electrically insulating layer.
10. The intravascular blood pump of claim 9, wherein the head portion of the cable comprises an adhesive layer beneath the plurality of conductive layers.
11. The intravascular blood pump of claim 9, wherein the head portion is O-shaped.
12. The intravascular blood pump of claim 9, wherein the head portion is C-shaped.
13. The intravascular blood pump of claim 9, wherein the tail portion comprises a coating covering sections of the plurality of electrically conductive layers.
14. The intravascular blood pump of claim 13, wherein the tail portion comprises one or more pads, and wherein the one or more pads of the tail portion and the one or more pads of the head portion are exposed.
15. The intravascular blood pump of claim 1, wherein the cable is bent near an interface between the tail portion and the head portion at an angle between 45 and 135 degrees.
16. A method for assembling an intravascular blood pump comprising a motor and a cable, the method comprising: coupling a head portion of the cable to an internal structure of the intravascular blood pump, wherein the cable further comprises a tail portion, wherein one or more electrical conduits extend through the tail and head portions, and wherein the head portion comprises one or more pads; and coupling at least one coil of the motor to at least one of the electrical conduits

through at least one of the pads, wherein the motor further comprises a rotor and a stator, wherein the stator comprises one or more coils, wherein the stator is configured to generate a rotating magnetic field in response to receiving one or more electrical signals at the one or more coils, and wherein the rotating magnetic field causes the rotor to rotate.

17. The method of claim 16, wherein the head portion is O-shaped or C-shaped.

18. The method of claim 17, wherein at least one of the electrical conduits comprises a plurality of electrically conductive layers and a plurality of through holes, and wherein each of the conductive layers is separated by at least one electrically insulating layer.

19. The method of claim 18, wherein the head portion comprises an adhesive layer beneath the plurality of conductive layers, and wherein coupling the head portion to the internal structure of the intravascular blood pump comprises placing the adhesive layer on the internal structure.

20. The method of claim 19, wherein coupling the at least one coil to the at least one of the electrical conduits through the at least one of the pads comprises soldering the at least one coil to the at least one of the pads.
